12.1 Precise Conditioning

WORKSHEET

The question

STEP I 2007, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Four discs are taken from eleven. Does the order of drawing matter for the event "all four are numbered"? How many equally likely sets of four discs are there?

STAGE 1 No condition

Covers: Part (i)

Ask yourself How many of the $\binom{11}{4}$ equally likely sets of four discs are all numbered?

STAGE 2 The 3 comes first

Covers: Parts (ii) and (iii)

Pause and try Once the disc numbered 3 has been drawn first, the other three discs are a random set of three from the ten left. How many of those ten are numbered? Count the sets you want.

STAGE 3 The 3 is taken (at some point)

Covers: Part (iv)

Ask yourself Does it matter when the 3 was drawn? Given that it is among the four, what can you say about the other three?

STAGE 4 A numbered disc comes first

Covers: Part (v)

Ask yourself Given that the first disc is numbered, whichever number it is, what are the other three discs?

STAGE 5 **A numbered disc is taken**

Covers: Part (vi)

Pause and try This condition is different: it does not tell you which disc is numbered. Count the sets of four with at least one numbered disc, and those with exactly two.

What's really going on

The boy-or-girl puzzle

Martin Gardner posed this in 1959: Mr Smith has two children, at least one of whom is a boy; what is the probability that both are boys? Listing the equally likely families BB, BG, GB, GG and removing GG leaves three, of which one is BB: the answer is $\frac{1}{3}$. But if we know that the *elder* child is a boy, the answer is $\frac{1}{2}$. These are parts (vi) and (v) of our question in miniature. Gardner later admitted that the question as he first posed it was ambiguous: it depends on how we came to know that there is a boy. If we met one of the children at random and he was a boy, the answer is $\frac{1}{2}$; if we asked "do you have at least one son?" and were told yes, it is $\frac{1}{3}$. Probability depends on the process that produced the information, not just the information.

Monty Hall

The same lesson underlies the most argued-about puzzle in probability. On a game show, a car is behind one of three doors. You pick door 1. The host, who knows where the car is, always opens one of the other two doors to reveal a goat, and always offers you the chance to switch; when the car is behind your door, so that he has a choice, he picks between the other two at random. He opens door 3, and offers you door 2. Should you switch? Yes: switching wins with probability $\frac{2}{3}$. The host's choice of door is not random; it depends on where the car is, and that carries information. The precise rules matter. If the host always opens door 3 whenever he can, then seeing door 3 open makes doors 1 and 2 equally likely, and switching wins with probability only $\frac{1}{2}$; yet a contestant who decides in advance always to switch still wins $\frac{2}{3}$ of the time, because switching wins exactly when the first choice was wrong. When Marilyn vos Savant gave this answer in a magazine column in 1990, she received thousands of letters telling

her she was wrong, some from mathematicians. A computer simulation, or a careful list of the equally likely cases, settles it.

Bayes and Laplace

Conditional probability was put on a systematic footing by Thomas Bayes, whose essay was published after his death in 1763, and by Pierre-Simon Laplace, who in 1774 stated the rule now called Bayes' theorem in its general form. Andrey Kolmogorov's axioms of 1933, which every university course now starts from, define $\mathrm{P}(A \mid B) = \mathrm{P}(A \cap B)/\mathrm{P}(B)$, and the rest of probability theory is built on that definition. The next section, Section 12.2, shows what Bayes' theorem does in practice.

Try this next

1. **Generalise.** With b blank and n numbered discs, and k drawn, show that the answers to (iv) and (v) are always equal, and find when (vi) equals them.

2. **Boys born on Tuesday.** A family has two children, and at least one is a boy born on a Tuesday. Show that the probability that both are boys is $\frac{13}{27}$. (List the 14×14 equally likely possibilities, or count cleverly.)

3. **Monty Hall.** Verify the $\frac{2}{3}$ by listing outcomes. What if the host does not know where the car is, opens one of the other doors at random, and it happens to show a goat?

4. **Two aces.** A bridge hand of 13 cards contains at least one ace. What is the probability that it contains at least two? Compare with the probability given that it contains the ace of spades.

Further reading

- **Jason Rosenhouse, *The Monty Hall Problem*** (Oxford University Press). The puzzle, its history and its many variations.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), the chapters on conditional probability.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online). Excellent on conditioning, with many paradoxes.

The full guided solution is in Section 12.1:
https://beyondstep.keithmcnulty.org/ch12/conditioning.html